

1

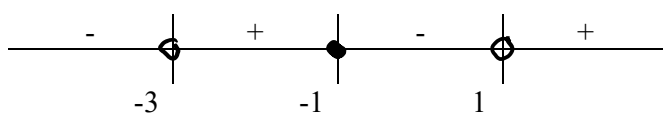
(i)

$$\frac{x+2}{x-1} \leq \frac{x}{x+3}$$

$$\frac{(x+2)(x+3) - x(x-1)}{(x-1)(x+3)} \leq 0$$

$$\frac{x^2 + 5x + 6 - x^2 + x}{(x-1)(x+3)} \leq 0$$

$$\frac{6(x+1)}{(x-1)(x+3)} \leq 0$$



$$x < -3 \text{ or } -1 \leq x < 1$$

(ii)

$$\frac{e^x + 2}{e^x - 1} \leq \frac{e^x}{e^x + 3}$$

Replace x with e^x ,

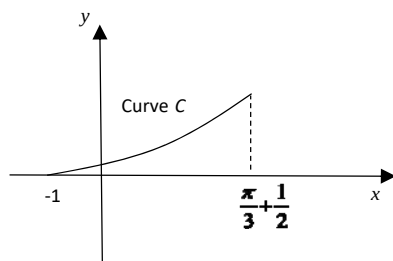
$$e^x < -3 \text{ (N.A.) or } -1 \leq e^x < 1$$

$$\Rightarrow e^x < 1$$

$$\Rightarrow x < 0$$

$$\therefore \{x \in \mathbb{R} : x < 0\}$$

2



$$x = t - \cos 2t$$

$$\frac{dx}{dt} = 1 + 2 \sin 2t$$

$$\text{Area} = \int_{-1}^{\frac{\pi}{3} + \frac{1}{2}} y dx$$

	$= \int_0^{\frac{\pi}{3}} \sin^2 t (1 + 2 \sin 2t) dt$ $= \int_0^{\frac{\pi}{3}} (\sin^2 t + 2 \sin^2 t \sin 2t) dt \text{ (Shown)}$ $\int_0^{\frac{\pi}{3}} (\sin^2 t + 2 \sin^2 t \sin 2t) dt$ $= \int_0^{\frac{\pi}{3}} \sin^2 t dt + 2 \int_0^{\frac{\pi}{3}} \sin^2 t \sin 2t dt$ $= \int_0^{\frac{\pi}{3}} \frac{1 - \cos 2t}{2} dt + 2 \int_0^{\frac{\pi}{3}} \sin^2 t (2 \cos t \sin t) dt$ $= \left[\frac{t}{2} - \frac{\sin 2t}{4} \right]_0^{\frac{\pi}{3}} + 4 \int_0^{\frac{\pi}{3}} \cos t (\sin^3 t) dt$ $= \frac{\pi}{6} - \frac{1}{4} \sin \frac{2\pi}{3} + 4 \left[\frac{\sin^4 t}{4} \right]_0^{\frac{\pi}{3}}$ $= \frac{\pi}{6} - \frac{1}{4} \left(\frac{\sqrt{3}}{2} \right) + \left(\frac{\sqrt{3}}{2} \right)^4$ $= \frac{\pi}{6} + \frac{9 - 2\sqrt{3}}{16}$
3	(a) Method 1 $\frac{3}{2} \int_0^b \frac{1}{\sqrt{\left(\frac{1}{2}\right)^2 - x^2}} dx = \frac{\pi}{2}$ $\left[\frac{3}{2} \sin^{-1} \left(\frac{x}{\frac{1}{2}} \right) \right]_0^b = \frac{1}{2} \pi$ $\left[\sin^{-1} 2x \right]_0^b = \frac{1}{3} \pi$ $\sin^{-1} 2b = \frac{1}{3} \pi$ $2b = \sin \frac{\pi}{3}$ $b = \frac{\sqrt{3}}{4}$ Method 2 $\frac{3}{2} \int_0^b \frac{2}{\sqrt{1 - (2x)^2}} dx = \frac{\pi}{2}$

$$\left[\sin^{-1} 2x \right]_0^b = \frac{1}{3} \pi$$

$$\sin^{-1} 2b = \frac{1}{3} \pi$$

$$2b = \sin \frac{\pi}{3}$$

$$b = \frac{\sqrt{3}}{4}$$

(b)

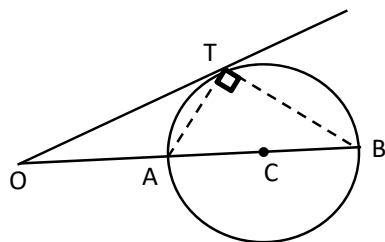
$$\int \frac{1}{\sqrt{4+x^2}} dx$$

$$\text{Let } x = 2 \tan \theta \Rightarrow \frac{dx}{d\theta} = 2 \sec^2 \theta$$

$$\begin{aligned} \int \frac{1}{\sqrt{4+x^2}} dx &= \int \frac{1}{\sqrt{4+4 \tan^2 \theta}} (2 \sec^2 \theta) d\theta \\ &= \int \frac{1}{2 \sec \theta} (2 \sec^2 \theta) d\theta \\ &= \int \sec \theta d\theta \\ &= \ln |\sec \theta + \tan \theta| + C \\ &= \ln \left| \sqrt{1 + \frac{x^2}{4}} + \frac{x}{2} \right| + C \end{aligned}$$

4

(i)



AT is perpendicular to *TB*.

$$(\mathbf{t} - \mathbf{a}) \cdot (\mathbf{t} - \mathbf{b}) = 0$$

$$(\mathbf{t} - \mathbf{a}) \cdot (\mathbf{t} - k\mathbf{a}) = 0 \quad (\text{Since } \mathbf{b} = k\mathbf{a})$$

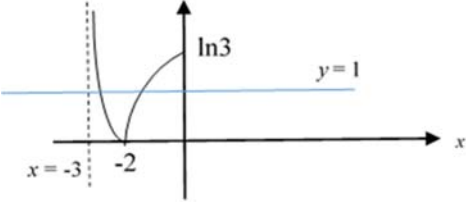
$$|\mathbf{t}|^2 - \mathbf{a} \cdot \mathbf{t} - k\mathbf{a} \cdot \mathbf{t} + k|\mathbf{a}|^2 = 0$$

$$\mathbf{a} \cdot \mathbf{t} = \frac{k|\mathbf{a}|^2 + |\mathbf{t}|^2}{k+1}$$

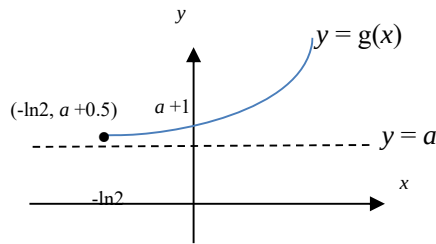
(ii)

$\angle OTC$ is 90° .

	$\Rightarrow \overrightarrow{OT} \cdot \overrightarrow{TC} = 0$ $\mathbf{t} \cdot (\mathbf{c} - \mathbf{t}) = 0$ Using Ratio Theorem, $\overrightarrow{OC} = \frac{\overrightarrow{OA} + \overrightarrow{OB}}{2} = \frac{\mathbf{a} + k\mathbf{a}}{2} = \frac{(k+1)\mathbf{a}}{2}$ $\mathbf{t} \cdot \left(\frac{(k+1)\mathbf{a}}{2} - \mathbf{t} \right) = 0$ $\frac{(k+1)}{2} \mathbf{a} \cdot \mathbf{t} = \mathbf{t} ^2$ $\frac{(k+1)}{2} \left(\frac{k \mathbf{a} ^2 + \mathbf{t} ^2}{k+1} \right) = \mathbf{t} ^2$ $k \mathbf{a} ^2 + \mathbf{t} ^2 = 2 \mathbf{t} ^2$ $k \mathbf{a} ^2 = \mathbf{t} ^2$ $\frac{ \mathbf{a} }{ \mathbf{t} } = \frac{1}{\sqrt{k}} \Rightarrow \mathbf{a} : \mathbf{t} = 1 : \sqrt{k}$
5	(i) $u_1 = 3, u_2 = 9, u_3 = 18.$ (ii) Given that $u_{r+1} = u_r + ar + b$ and $u_1 = 3, u_2 = 9, u_3 = 18$, we have $u_2 = u_1 + a + b$ and $u_3 = u_2 + 2a + b$. $\Rightarrow 9 = 3 + a + b$ and $18 = 9 + 2a + b$ Solving the equations, $a = 3, b = 3$. (Shown) (iii) $\sum_{r=1}^{n-1} (3r + 3) = 6 + 9 + 12 + \dots + (3(n-1) + 3)$ $= \frac{n-1}{2} [6 + 3n]$ $= \frac{3(n-1)(n+2)}{2}$ $u_{r+1} - u_r = 3r + 3$ $\Rightarrow \sum_{r=1}^{n-1} (u_{r+1} - u_r) = \sum_{r=1}^{n-1} (3r + 3)$

	$\sum_{r=1}^{n-1} (3r+3) = \sum_{r=1}^{n-1} (u_{r+1} - u_r)$ $= \cancel{u_2} - u_1$ $+ \cancel{u_3} - \cancel{u_2}$ $+ \cancel{u_4} - \cancel{u_3}$ $+ \dots$ $+ u_n - \cancel{u_{n-1}}$ $= u_n - u_1$ $u_n - u_1 = \frac{3(n-1)(n+2)}{2} \text{ -----} (*)$ $u_n = \frac{3(n-1)(n+2)}{2} + u_1$ $= \frac{3(n-1)(n+2)}{2} + 3$ $= \frac{3(n^2 + n - 2 + 2)}{2}$ $= \frac{3n(n+1)}{2}$
6	(i)  The horizontal line $y = 1$ that cuts the graph $y = f(x)$ at more than one point. Thus f is not one-one function, f^{-1} does not exist. (ii) -2 (iii) $y = \ln(x+3) $ Since $-3 < x \leq -2$, $\ln(x+3) < 0$, $\therefore y = -\ln(x+3)$. $e^{-y} = x+3$ $x = e^{-y} - 3$ $f^{-1}: x \mapsto e^{-x} - 3, \quad x \geq 0$ (iv)

gf exists $\Leftrightarrow R_f \subseteq D_g$
 $R_f = [0, \infty) \subseteq D_g = [-\ln 2, \infty)$
 Thus gf exists.



Range of gf:
 $(-3, -2] \xrightarrow{f} [0, \infty) \xrightarrow{g} [a+1, \infty)$
 $R_{gf} = [a+1, \infty)$

7

(i)

Let $u = (\ln x)^2$ $v' = 1$

$$u' = 2(\ln x) \frac{1}{x} \quad v = x$$

$$\int (\ln x)^2 dx = x(\ln x)^2 - 2 \int \ln x dx$$

$$\int \ln x dx$$

$$= x \ln x - \int 1 dx \quad u = \ln x \quad v' = 1$$

$$= x \ln x - x + C \quad u' = \frac{1}{x} \quad v = x$$

$$\int (\ln x)^2 dx = x(\ln x)^2 - 2x \ln x + 2x + C$$

(ii)

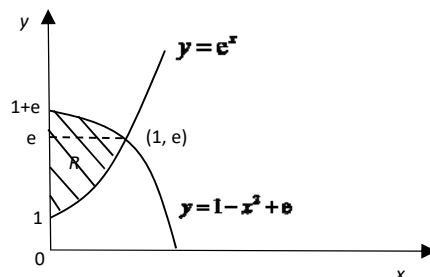
When $x=1$, $y=e^1=e$

When $x=1$, $y=1+e-(1)^2=e$

$\therefore (1, e)$ is the point of intersection of the curves $y=e^x$ and $y=1-x^2+e$

$$y=1-x^2+e$$

$$x^2=1+e-y$$



	$ \begin{aligned} V &= \pi \int_e^{1+e} x^2 dy + \pi \int_1^e (\ln y)^2 dy \\ &= \pi \int_e^{1+e} 1 + e - y dy + \pi \left[y(\ln y)^2 - 2y \ln y + 2y \right]_1^e \\ &= \pi \left[y + ey - \frac{y^2}{2} \right]_e^{1+e} + \pi \left[y(\ln y)^2 - 2y \ln y + 2y \right]_1^e \\ &= \pi \left([1+e+e(1+e) - \frac{(1+e)^2}{2}] - [e+e^2 - \frac{e^2}{2}] \right) + \pi [e - 2e + 2e - 2] \\ &= \pi \left[\frac{1}{2} \right] + \pi [e - 2] \\ &= \pi \left(e - \frac{3}{2} \right) \end{aligned} $
8	(i) $\sin \theta = \frac{\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}}{\sqrt{2}\sqrt{6}} = \frac{1}{\sqrt{2}\sqrt{6}}$ $\theta = 16.8^\circ$ (1 d.pl.) (ii) Let C be the point of intersection of line l_1 and plane. $\therefore \overrightarrow{OC} = \begin{pmatrix} 2 \\ 2 + \lambda \\ 3 + \lambda \end{pmatrix} \text{ for some value of } \lambda.$ Since C is also on the plane Π, $\begin{pmatrix} 2 \\ 2 + \lambda \\ 3 + \lambda \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = 4$ $2 + 4 + 2\lambda - 3 - \lambda = 4$ $\lambda = 1$ $\overrightarrow{OC} = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}$ (iii) Method 1 The line parallel to normal and passing through A is

$$\mathbf{r} = \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix} + \alpha \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}, \quad \alpha \in \mathbb{R}.$$

$$\begin{pmatrix} 2+\alpha \\ 2+2\alpha \\ 3-\alpha \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = 4$$

$$2 + \alpha + 4 + 4\alpha - 3 + \alpha = 4$$

$$\alpha = \frac{1}{6}$$

$$\overrightarrow{OF} = \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix} + \frac{1}{6} \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = \frac{1}{6} \begin{pmatrix} 13 \\ 14 \\ 17 \end{pmatrix}$$

Method 2 (Finding projection vector)

$$\overrightarrow{AC} = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

Let F be foot of perpendicular from A to Π .

$$\overrightarrow{AF} = \left(\overrightarrow{AC} \cdot \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \right) \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

$$= \frac{\left(\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \right) \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}}{\sqrt{6}\sqrt{6}}$$

$$= \frac{1}{6} \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

$$\overrightarrow{OF} = \overrightarrow{OA} + \overrightarrow{AF}$$

$$= \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix} + \frac{1}{6} \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = \frac{1}{6} \begin{pmatrix} 13 \\ 14 \\ 17 \end{pmatrix}$$

(iii)

To find $\overrightarrow{OA'}$:

Method 1 (using parallel vectors)

$$\overrightarrow{AF} = \frac{1}{6} \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

$$\overrightarrow{AA'} = 2\overrightarrow{AF} = \frac{1}{3} \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

$$\overrightarrow{OA'} = \overrightarrow{OA} + \overrightarrow{AA'} = \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix} + \frac{1}{3} \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 7 \\ 8 \\ 8 \end{pmatrix}$$

Method 2 (using equal vectors)

$$\overrightarrow{AF} = \frac{1}{6} \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

$$\overrightarrow{FA'} = \overrightarrow{AF} = \frac{1}{6} \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

$$\overrightarrow{OA'} = \overrightarrow{OF} + \overrightarrow{FA'} = \frac{1}{6} \begin{pmatrix} 13 \\ 14 \\ 17 \end{pmatrix} + \frac{1}{6} \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 7 \\ 8 \\ 8 \end{pmatrix}$$

Method 3 (using Ratio Theorem)

$$\overrightarrow{OF} = \frac{1}{2}(\overrightarrow{OA} + \overrightarrow{OA'})$$

$$\overrightarrow{OA'} = 2\overrightarrow{OF} - \overrightarrow{OA} = \frac{1}{3} \begin{pmatrix} 13 \\ 14 \\ 17 \end{pmatrix} - \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 7 \\ 8 \\ 8 \end{pmatrix}$$

$$\text{Thus } \overrightarrow{A'C} = \overrightarrow{OC} - \overrightarrow{OA'} = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} - \frac{1}{3} \begin{pmatrix} 7 \\ 8 \\ 8 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} -1 \\ 1 \\ 4 \end{pmatrix}$$

Thus equation of reflection line is

$$\mathbf{r} = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} + \beta \begin{pmatrix} -1 \\ 1 \\ 4 \end{pmatrix}, \quad \beta \in \mathbb{R}.$$

Cartesian equation of reflection line is

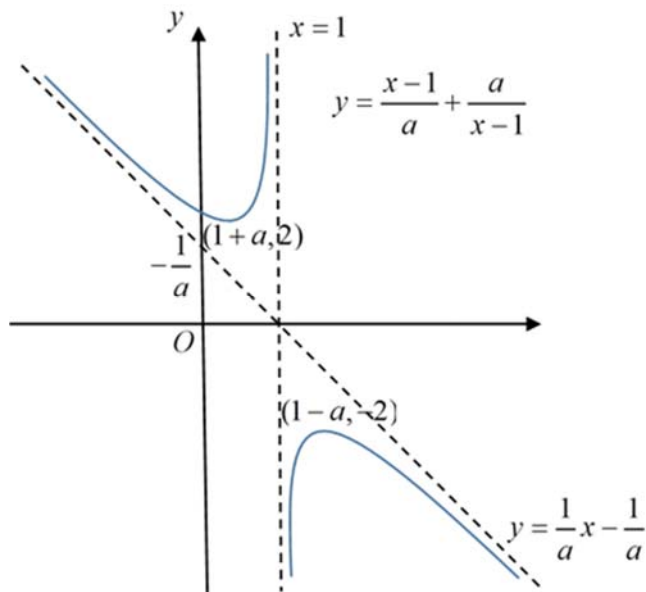
$$\frac{x-2}{-1} = \frac{y-3}{1} = \frac{z-4}{4}.$$

(iv)

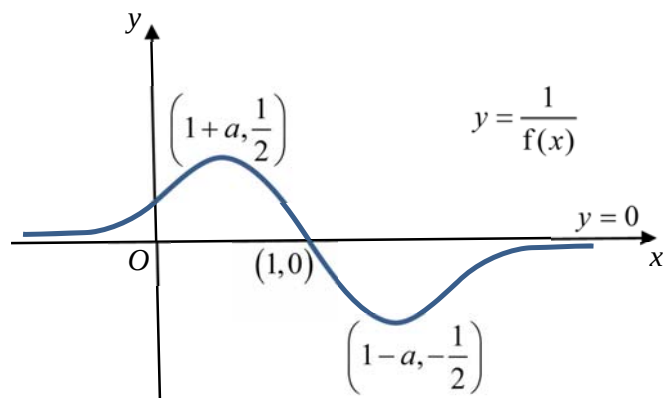
$$l_1 : \mathbf{r} = \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad \lambda \in \mathbb{R}$$

	$l_2 : \mathbf{r} = \begin{pmatrix} -1 \\ 4 \\ 7 \end{pmatrix} + \mu \begin{pmatrix} -3 \\ 1 \\ 1 \end{pmatrix}, \quad \mu \in \mathbb{R}$ $\begin{pmatrix} 2 \\ 2+\lambda \\ 3+\lambda \end{pmatrix} = \begin{pmatrix} -1-3\mu \\ 4+\mu \\ 7+\mu \end{pmatrix}$ $\Rightarrow \begin{cases} 2 = -1-3\mu \\ 2+\lambda = 4+\mu \\ 3+\lambda = 7+\mu \end{cases}$ $\Rightarrow \text{No solution.}$ Since the lines are non-parallel as $\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \neq k \begin{pmatrix} -3 \\ 1 \\ 1 \end{pmatrix}$ for all $k \in \mathbb{R}$ and do not intersect, thus they are skew lines.
9	(i) Let $y = \frac{x-1}{a} + \frac{a}{x-1}$ $\Rightarrow ay(x-1) = (x-1)(x-1) + a^2$ $\Rightarrow x^2 - (2+ay)x + 1 + a^2 + ay = 0$ Since x is real, discriminant $= (2+ay)^2 - 4(1)(1+a^2+ay) \geq 0$ $\Rightarrow 4 + 4ay + a^2y^2 - 4 - 4a^2 - 4ay \geq 0$ $\Rightarrow a^2y^2 - 4a^2 \geq 0$ $\Rightarrow y^2 - 4 \geq 0$ $\Rightarrow (y-2)(y+2) \geq 0$ $\Rightarrow y \leq -2 \text{ or } y \geq 2$ The set of values that $f(x)$ can take is $\{y : y \in \mathbb{R}, y \leq -2 \text{ or } y \geq 2\}$ (ii) $f(x) = \frac{x-1}{a} + \frac{a}{x-1}$ $f'(x) = \frac{1}{a} + a \left[-(x-1)^{-2} \right]$ At stationary points, $f'(x) = \frac{1}{a} - \frac{a}{(x-1)^2} = 0$ $\Rightarrow (x-1)^2 = a^2$ $\Rightarrow x = 1 \pm a$ When $x = 1 - a, y = -2$ When $x = 1 + a, y = 2$ Equations of asymptotes:

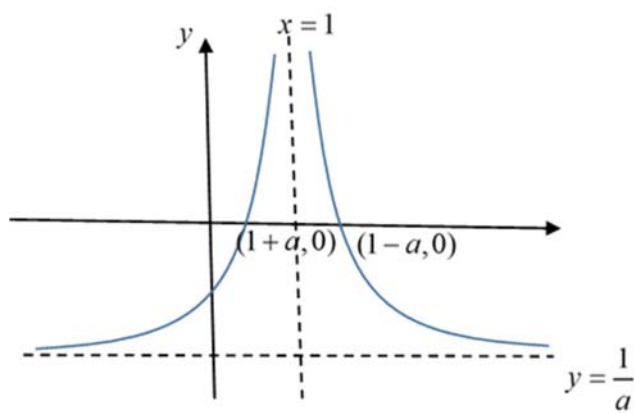
$$x=1, y=\frac{1}{a}x-\frac{1}{a}.$$



(iiiia)

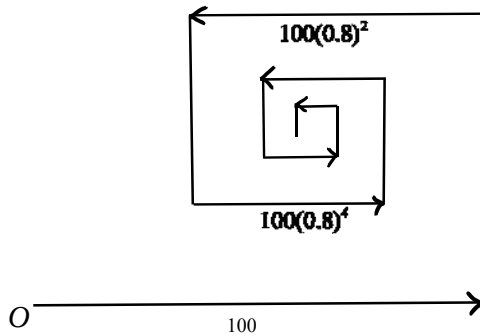


(iiib)



(iv)

	$y = \frac{x-1}{a} + \frac{a}{x-1}$ Let $g(x) = \frac{x-1}{a} + \frac{a}{x-1} \xrightarrow{x \rightarrow x+1} g(x+1) = \frac{x}{a} + \frac{a}{x}$ $\xrightarrow{y \rightarrow 2y} 2g(x+1) = \frac{x}{a} + \frac{a}{x} \Rightarrow g(x+1) = \frac{x}{2a} + \frac{a}{2x}$ The curve C is translated in the negative x-direction by 1 unit, followed by scaling parallel to y-axis by a scale factor of $\frac{1}{2}$. <u>OR</u> The curve C is scaled parallel to the y-axis by a scale factor of $\frac{1}{2}$, followed by a translation in the negative x-direction by 1 unit.												
10	(i) <table border="1"> <thead> <tr> <th>Between $(n-1)^{\text{th}}$ & n^{th} turn</th><th>Distance covered during the n^{th} turn</th></tr> </thead> <tbody> <tr> <td>1</td><td>100</td></tr> <tr> <td>2</td><td>$100(0.8)$</td></tr> <tr> <td>3</td><td>$100(0.8)^2$</td></tr> <tr> <td>4</td><td>$100(0.8)^3$</td></tr> <tr> <td>n</td><td>$100(0.8)^{n-1}$</td></tr> </tbody> </table> Distance covered between its 11^{th} and 12^{th} turn $= 1000(0.8)^{11} = 8.59 \text{ m}$ (ii) Let the required number of turns be n. $100 + 100(0.8) + 100(0.8)^2 + \dots + 100(0.8)^{n-1} = 485$ $\frac{100[1 - 0.8^n]}{1 - 0.8} = 485$ $1 - 0.8^n = 0.97$ $0.8^n = 0.03$ $n = \frac{\ln 0.03}{\ln 0.8}$ $n = 15.7$ The robot has made 15 turns after covering a total distance of 485 m. (iii)	Between $(n-1)^{\text{th}}$ & n^{th} turn	Distance covered during the n^{th} turn	1	100	2	$100(0.8)$	3	$100(0.8)^2$	4	$100(0.8)^3$	n	$100(0.8)^{n-1}$
Between $(n-1)^{\text{th}}$ & n^{th} turn	Distance covered during the n^{th} turn												
1	100												
2	$100(0.8)$												
3	$100(0.8)^2$												
4	$100(0.8)^3$												
n	$100(0.8)^{n-1}$												



Eventual horizontal displacement

$$= 100 - 100(0.8)^2 + 100(0.8)^4 - 100(0.8)^6 + \dots$$

$$= \frac{100}{1 - [-(0.8)^2]}$$

$$= 60.976 \approx 61.0 \text{ m}$$

Eventual vertical displacement

$$= 100(0.8) - 100(0.8)^3 + 100(0.8)^5 - \dots$$

$$= \frac{100(0.8)}{1 - [-(0.8)^2]} \approx 48.8 \text{ m}$$

It will end up 61.0 m due east and 48.8 m due north of O .

Total distance covered after 8 turns

$$= 100 + 100\left(1 - \frac{x}{100}\right) + 100\left(1 - \frac{x}{100}\right)^2 + \dots + 100\left(1 - \frac{x}{100}\right)^8$$

$$= \frac{100(1 - r^9)}{1 - r} \quad \text{where } r = 1 - \frac{x}{100}$$

$$\therefore \frac{100(1 - r^9)}{1 - r} = 500$$

$$\Rightarrow r^9 - 5r + 4 = 0 \text{ --- (1)}$$

$$r = -1.30 \text{ (NA)}, 1 \text{ (NA)}, 0.84300178$$

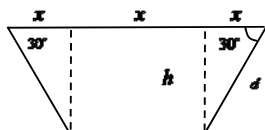
$$\text{Thus, } 1 - \frac{x}{100} = 0.84300178$$

$$\Rightarrow x = 15.7$$

11

(i)

Let h be the height of the trapezium, and d be the diagonal length of trapezium.



$$\frac{x}{d} = \cos 30^\circ$$

$$\Rightarrow d = \frac{2}{\sqrt{3}}x$$

We have $\tan 30^\circ = \frac{h}{x}$

$$h = \frac{x}{\sqrt{3}}$$

$$\text{Volume of the water trough} = \frac{(3+1)x^2y}{2} \tan 30^\circ$$

$$= \frac{2x^2y}{\sqrt{3}} = \frac{2}{\sqrt{3}}x^2y$$

$$\therefore \frac{20}{\sqrt{3}} = \frac{2x^2y}{\sqrt{3}}$$

$$y = \frac{10}{x^2}$$

$$A = xy + 2(d)(y) + 2\left(\frac{3x+x}{2}\right)(h)$$

$$= x\left(\frac{10}{x^2}\right) + 2\left(\frac{2x}{\sqrt{3}}\right)\left(\frac{10}{x^2}\right) + 2\left(2x\left(\frac{x}{\sqrt{3}}\right)\right)$$

$$= \frac{10}{x} + \frac{40}{x\sqrt{3}} + \frac{4x^2}{\sqrt{3}}$$

$$= \frac{4x^2}{\sqrt{3}} + \left(\frac{40\sqrt{3} + 30}{3x}\right) \text{ (Shown)}$$

(ii)

$$\frac{dA}{dx} = \frac{8}{\sqrt{3}}x - \left(\frac{40\sqrt{3} + 30}{3}\right)x^{-2} = 0$$

$$x^3 = \frac{(40\sqrt{3} + 30)\sqrt{3}}{24}$$

$$x = 1.93 \text{ m}$$

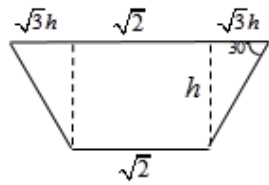
$$\frac{d^2A}{dx^2} = \frac{8}{\sqrt{3}} + 2\left(\frac{40\sqrt{3} + 30}{3}\right)x^{-3} > 0 \text{ when } x = 1.93$$

$x = 1.93 \text{ m}$ when A is minimum.

(iii)

When $y = 5$, $\frac{2x^2(5)}{\sqrt{3}} = \frac{20}{\sqrt{3}}$

$$\Rightarrow x^2 = 2 \Rightarrow x = \sqrt{2}$$



$\therefore$ Volume of liquid feed V

$$= \frac{(2\sqrt{2} + 2\sqrt{3}h)}{2} \times 5 \times h$$

$$= 5(\sqrt{2} + \sqrt{3}h)h = 5\sqrt{2}h + 5\sqrt{3}h^2$$

$$\therefore \frac{dV}{dh} = 5\sqrt{2} + 10\sqrt{3}h$$

$$\text{When } t = 72, \text{ Volume} = \frac{20}{\sqrt{3}} - 0.72$$

$$\therefore 5\sqrt{2}h + 5\sqrt{3}h^2 = \frac{20}{\sqrt{3}} - 0.72$$

$$\Rightarrow h = 0.7820717$$

$$\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$$

$$-0.01 = (5\sqrt{2} + 10\sqrt{3})(0.7820717) \times \frac{dh}{dt}$$

$$\frac{dh}{dt} = -0.000485 \text{ m/hr}$$

The rate at which the depth of the water trough is decreasing after 72 hours is 0.000485 m/hr